

Experiment Design for Computer Sciences (0AL0400)

Paper Study – "Benchmarking in Optimization: Best Practice and Open Issues" Part II

Claus Aranha
caranha@cs.tsukuba.ac.jp

University of Tsukuba, Department of Computer Sciences

Version 2024.1 (Updated June 21, 2024)

Outline

We continue the study of the "Benchmarking Best Practices" paper. The topics for the second half are:

- Analyzing the results;
- Selecting of experimental parameters;
- Reporting the Results;

<https://arxiv.org/abs/2007.03488>

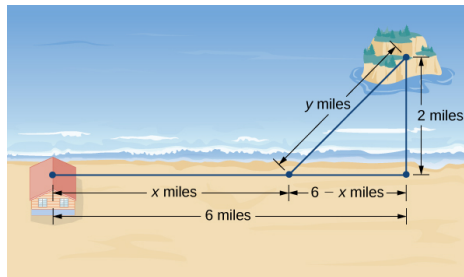
The paper focuses on the study of optimization methods, but its suggestions are highly applicable for other CS areas as well.

Review: What is an optimization problem?

Consider the following problem: You want to leave from your home, and reach a beautiful island. Your walking speed is 8mph, and your swimming speed is 3mph.

What is the optimal (fastest) location for you to switch from walking to swimming?

This problem can be represented as an equation, and you can find the answer by calculating the value of x that minimizes $T(x)$



Equation for the beach problem:

$$T(x) = \frac{x}{8} + \frac{\sqrt{(6-x)^2 + 4}}{3}$$

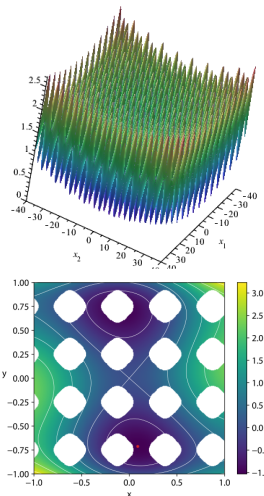
Optimization Benchmark Problems

Most optimization problems are much more complicated. Benchmark functions can replicate these conditions:

- Griewank benchmark problem (2 dimensions);
$$G(x) = 1 + \frac{1}{4000}x_1^2 + \frac{1}{4000}x_2^2 - \cos(x_1) \cos(\frac{1}{2}x_2\sqrt{2})$$
- Gomez and Levy problem (modified);
$$GL(x, y) = 4x^2 - 2.1x^4 + \frac{1}{3}x^6 + xy - 4y^2 + 4y^4$$

subjected to: $-\sin(4\pi x) + 2\sin^2(2\pi y) \leq 1.5$

There are large sets of benchmark problems like this, such as the CEC, BBOB, etc.



Optimization Algorithms

Optimization algorithms try to find the value of x (vector) that minimize (or maximize) the problem function.

There are many different optimization algorithms, and each algorithm can be further configured with different parameters:

- Genetic Algorithm;
- Evolution Strategies;
- Particle Swarm Optimization;
- CMA-ES;
- and many others...

We will focus on **Metaheuristic** optimization algorithms, which work by testing several candidate values for x until they find the optimum.

Three-Level Approach for Result Analysis

In the first half we discussed the selection of goal, algorithms, datasets and measures for the experiment. After that, we perform the experiment and collect the data. **The next step is to analyze the data and draw conclusions** from it.

Consider a comparison experiment, between one or more algorithm configurations, on one or more problem instances.

We can divide the analysis in a three level approach:

- Exploratory Data Analysis (EDA)
- Confirmatory Analysis
- Relevance Analysis

Please check the many references for each of these levels in the original paper.

Exploratory Data Analysis (EDA)

Descriptive and graphical techniques for an initial understanding of the results:

- Verify the **distribution** of the data;
- Check underlying assumptions: **Normality, equality of variances**;
- **Visualize basic patterns** in the data: outliers, dependencies;
- Catch possible bugs in the implementation;
- "Let the data speak"

Exploratory Data Analysis is based on experience, judgement and artistry: there is no standard cookbook, but many recipes.

EDA: The Glorious Seven

Standard metrics that we want to calculate from the sample:

- Mean
- Median
- Minimum
- Maximum
- First Quartile
- Third Quartile
- Standard Deviation

These values may be **Sensitive to outliers, missing or biased data.**

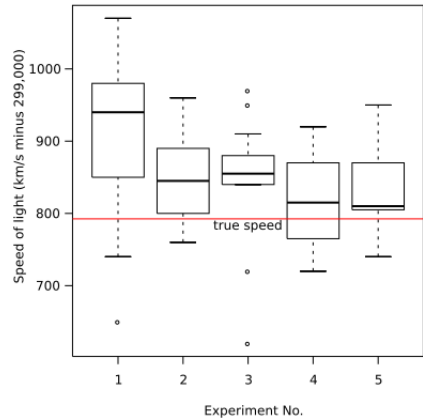
These values are important, but they are based on that specific experiment only, so **they do not provide a complete analysis of the performance.**

EDA: Graphical Tools – Boxplot

The Boxplot visualizes the **Median, First Quartile, Third Quartile** of a sample, as well as the 5 and 95 percentiles.

They can show the spread of a sample in a very compact way, and are also useful for visually comparing multiple samples.

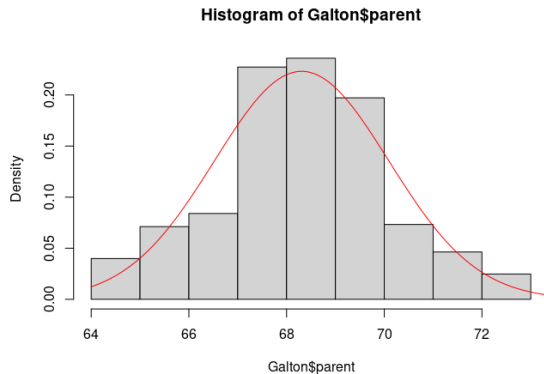
A boxplot that includes density information is a **Violin Plot**.



EDA: Graphical Tools – Histogram

The Histogram can show the shape of distribution of the data in the sample, and also give an idea of the statistical bias and spread of the data.

The histogram is very sensitive to the size of the boxes (*bins*), so it is suggested to combine histograms with density plots.



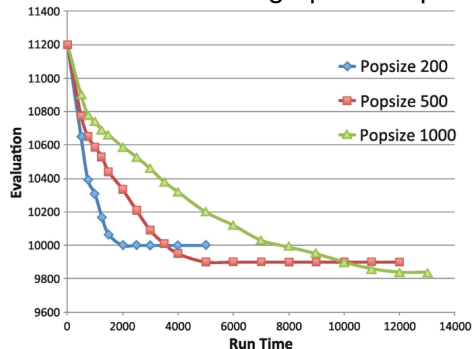
EDA: Graphical Tools – Experiment Specific Tools

In addition to graphs that show aggregate results of the experiment, optimization research often uses specific graphs that show the behavior of algorithms in detail, such as:

- **Run-time Behavior**
- **Performance Profiles**
- **Data Profiles**

Runtime behavior is the most commonly used for comparing metaheuristic algorithms.

Run-time behavior graph example



Confirmatory Analysis

After the EDA step, the Confirmatory Analysis level uses **Inferential Statistical Tests** to verify differences of smaller magnitude when comparing different algorithm configurations.

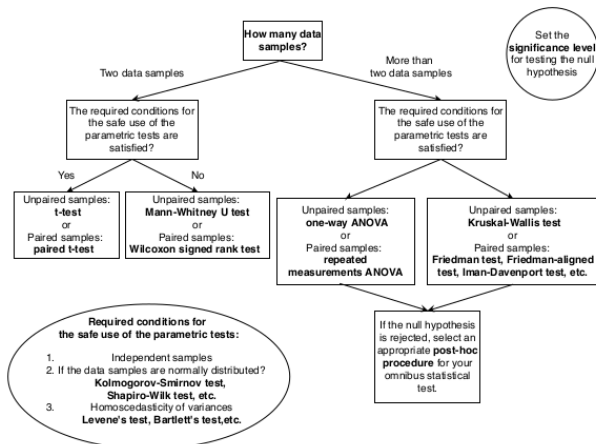
In this stage, the characteristics of the experiment should be analyzed to choose the most appropriate statistical test:

- Number of data samples;
- Conditions for using parametric tests;
(independence, normality, homoscedasticity of variances)
- Verification if the data is paired or unpaired;

The choice and procedure of the test follow the ideas already described earlier in this course.

Confirmatory Analysis – Pipeline for selecting statistical tests

Pipeline for Selecting Most Commonly used Statistical Tests



Relevance Analysis

After the statistical confirmatory analysis, the third step is to validate these results: "Are the differences meaningful in practice or only statistical artifacts"?

Assembly line

Two optimization algorithms to minimize the time of a production process are compared. The mean difference in performance is $\delta = 10^{-14}$ which is statistically relevant. However, this difference has no meaning in reality, because it is below the precision of the timer.

The performance measures in benchmark studies can be affected by factors that are irrelevant in the final application, such as: computer accuracy (floating point), variable types, and stopping criteria.

Relevance Analysis: Severity Measure

Severity is a measure that can be used to determine the meaningfulness of a statistically significant result.

Severity is calculated as the power of a post-hoc data analysis that evaluates how well the data fits the testing framework of the original hypothesis.

The calculation of the severity measure deals directly with the **problem of large n** that can happen in simulated benchmarking experiments.

Relevance Analysis: Multiple-Problem Analysis

When comparing the results over multiple problems (benchmark sets), it is very important to define a **draw threshold**: A value where even statistically significant differences are considered to be equal from an algorithm comparison point of view.

Using the draw threshold, recent techniques have been developed to evaluate the practical relevance of a comparison, such as "Chess Rating Systems" and "practical Deep Statistical Comparison".

In practice, there are still several open questions regarding the evaluation of problem relevance in benchmark studies.

Experimental Design

In experiments comparing algorithm configurations, "experimental design" usually refers to how we choose the parameters of the configurations that we want to compare. Some key concepts:

- **Design Space**: The combination of all variables under consideration;
- **Sequential design**: The choice of parameters happens after each result is investigated;
- **One Factor At a Time**: Sequential design where each parameter is investigated in order, fixing all the others;
- **Factorial design**: All combinations of parameters are investigated;
- **Space filling design**: A representative set of parameters are investigated; (example: Latin Hypercube)

Selection of the design depends on the goals and characteristics of the experiment.

Presentation of Results

Recommendations from Gent and Walsh (1994)

General ideas for improving the production of the paper with the experiment Results.

- Present statistics when describing the comparison between two algorithms;
- Do not push deadlines: Plan your experiment, analysis and writing to have enough time to carefully analyse the results. Spend time planning the Exploratory Analysis as well.
- Report Negative results: Present and discuss problem instances where the algorithm performs badly or does not perform as expected (key component of a scientific report as opposed to a PR piece)

The section lists several other papers with good advice and principles for scientific reporting.

Considerations about Reproducibility: Levels of Reproducibility

- **Repeatability:** Same team, same experimental setup.
- **Reproducibility:** Different team, different experimental setup.
- **Replicability:** Different team, different experimental setup.

Final Considerations

- Read the paper for more details and further discussion;
- Example exam in manaba;
- Files for Report 2 are delayed. I hope they will be finished this weekend.
- I will change the grade to: $R_1 * 0.4 + \max(R_2, E) * 0.6$

About these Slides

These slides were made by Claus Aranha, 2022. You are welcome to copy, re-use and modify this material.

These slides are a modification of "Design and Analysis of Experiments (2018)" by Felipe Campelo, used with permission.

Individual images in some slides might have been made by other authors. Please see the following references for those cases.

Image Credits I

[Page 3] Beach Optimization Image from math.libretexts.org.

[Page 4] Griewank function by Gisling, Gomez-Levi function by Nschloe, CC BY-SA 4.0, Wikimedia foundation.

[Page 9] Example of Boxplot from Wikipedia – Boxplots

[Page 10] Example of Histogram/Density Plot from University of Iowa - STAT:4580
<https://homepage.divms.uiowa.edu/luke/classes/STAT4580/histdens.html>

[Page 11] Example of Runtime Behavior graph from "Next Generation Genetic Algorithms: A User's Guide and Tutorial", by Darrel Whitley, in "Handbook of Metaheuristics", page 245–274, 2018

[Page 13] Pipeline for selecting statistical tests from Eftimov et al. 2020 – check Benchmark paper for full reference